Question 1: What is Simple Linear Regression (SLR)? Explain its purpose.

Answer:
Simple Linear Regression (SLR) is a statistical method used to model the relationship between one independent variable (X) and one dependent variable (Y) using a straight line.

Purpose:

- To predict the value of Y based on X
- To understand the strength and direction of the relationship between variables

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Question 2: What are the key assumptions of Simple Linear Regression?

Answer:
The main assumptions are:

1. **Linearity** – Relationship between X and Y is linear
2. **Independence** – Observations are independent of each other
3. **Homoscedasticity** – Constant variance of errors
4. **Normality** – Errors are normally distributed
5. **No multicollinearity** – (mainly for multiple regression, but still relevant conceptually)

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Answer:
The equation is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- **Y** = Dependent variable
- **X** = Independent variable
- **β□ (intercept)** = Value of Y when X = 0
- **β□ (slope)** = Change in Y for a one-unit change in X
- **ε (error term)** = Random error

Question 4: Provide a real-world example where simple linear regression can be applied.

Answer:
Example: Predicting house prices based on size.

- X = Size of the house (square feet)
- Y = Price of the house

SLR helps estimate how much price increases with size.

Question 5: What is the method of least squares in linear regression?

Answer:
The **method of least squares** finds the best-fitting line by minimizing the sum of squared differences between actual and predicted values.

$$\text{Minimize } \sum(Y_i - \hat{Y}_i)^2$$

This ensures the regression line is as close as possible to the data points.

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Answer:
Logistic Regression is a classification algorithm used to predict categorical outcomes (e.g., yes/no, 0/1).

Differences:

- Linear Regression → Predicts continuous values
- Logistic Regression → Predicts probabilities (0 to 1)
- Linear uses a straight line
- Logistic uses a sigmoid (S-shaped) curve

Question 7: Name and briefly describe three common evaluation metrics for regression models.

Answer:

1. **Mean Absolute Error (MAE)**
    - Average absolute difference between actual and predicted values

2. **Mean Squared Error (MSE)**
    - Average of squared differences (penalizes large errors more)
3. **Root Mean Squared Error (RMSE)**
    - Square root of MSE (interpretable in same units as data)

Question 8: What is the purpose of the R-squared metric in regression analysis?

Answer:
R-squared measures how well the model explains the variability of the dependent variable.

- Value ranges from 0 to 1
- Higher value = better fit
- Example: $R^2 = 0.8$ means 80% of variance is explained by the model

Question 9: Python code to fit a simple linear regression model using scikit-learn

```
from sklearn.linear_model import LinearRegression

import numpy as np


# Sample data

X = np.array([[1], [2], [3], [4], [5]])

y = np.array([2, 4, 6, 8, 10])


# Create model

model = LinearRegression()


# Fit model

model.fit(X, y)


# Get slope and intercept

slope = model.coef_[0]
```

intercept = model.intercept_

print("Slope:", slope)

print("Intercept:", intercept)

output:

Slope: 2.0000000000000004

Intercept: -1.7763568394002505e-15

Question 10: How do you interpret the coefficients in a simple linear regression model?

Answer:

In a simple linear regression:

$$y = \beta_0 + \beta_1 x$$

- β_0**(intercept): Value of** y**when** $x = 0$**(starting point).**
- β_1**(slope): Change in** y**for a 1-unit increase in** x**(rate of change).**

In short: intercept = baseline, slope = effect of x**on** y**.**